



2.4" TFT Display without Touch (240x320)

👉 Introduction

The **2.4-inch SPI TFT Display Module with ST7789V** is a high-resolution, full-color LCD designed for embedded applications. Featuring a 240 × 320-pixel LCD screen, it offers wide viewing angles and vibrant color reproduction. The ST7789V driver IC enables efficient communication via SPI, making it compatible with STM32, ESP32, Raspberry Pi, and Arduino. With low power consumption, a high refresh rate, and an optional touchscreen, this module is ideal for graphical user interfaces (GUIs), real-time data visualization, and portable embedded systems.

👉 Feature

1. **Display Type:** 2.4-inch TFT LCD
2. **Resolution:** 240 × 320 pixels
3. **Driver IC:** ST7789V
4. **Interface:** SPI
5. **Data Format:** RGB565
6. **Touch Support:** No

👉 Arduino Hardware Connection

TFT Display	Arduino Pin
VCC	5V
GND	GND
RESET	8
SCK	13
DC	9
CS	10
SDI(MOSI)	11
LED	5V
SDO(MISO)	12

👉 Arduino Demo Code

```
#include <Adafruit_GFX.h>    // Core graphics library
#include <Adafruit_ST7789.h> // Hardware-specific library for ST7789
#include <SPI.h>

// Define pins for the ST7789
#define TFT_MOSI 11 // Data out
#define TFT_SCLK 13 // Clock out
#define TFT_CS   10 // Chip select
#define TFT_RST   8  // Reset pin
#define TFT_DC    9  // Data/Command pin

// Color definitions
#define DISPLAY_COLOR_BLACK  0x0000
#define DISPLAY_COLOR_BLUE   0x001F
#define DISPLAY_COLOR_RED     0xF800
#define DISPLAY_COLOR_GREEN   0x07E0
#define DISPLAY_COLOR_CYAN    0x07FF
#define DISPLAY_COLOR_MAGENTA 0xF81F
#define DISPLAY_COLOR_YELLOW  0xFFE0
#define DISPLAY_COLOR_WHITE   0xFFFF

// Initialize ST7789 display
Adafruit_ST7789 tft = Adafruit_ST7789(TFT_CS, TFT_DC, TFT_MOSI, TFT_SCLK,
TFT_RST);

// Timer variables for toggling display
unsigned long lastToggleTime = 0;
const unsigned long toggleInterval = 3000; // 3 seconds

void setup() {
  // Initialize Serial for debugging
  Serial.begin(115200);
  while (!Serial);

  // Initialize the ST7789 display
  tft.init(240, 320); // 240x320 resolution
  tft.setRotation(1); // Adjust orientation if needed
  tft.fillScreen(DISPLAY_COLOR_BLACK); // Clear the screen

  // Demo initialization
  Serial.println("Display initialized. Starting demo...");
  demo();
}

void loop() {
  // Toggle display on/off every 3 seconds
```

```

if (millis() - lastToggleTime >= toggleInterval) {
    lastToggleTime = millis();
    static bool displayOn = true;

    displayOn = !displayOn;
    tft.enableDisplay(displayOn);

    if (displayOn) {
        Serial.println("Display turned ON.");
        demo();
    } else {
        Serial.println("Display turned OFF.");
    }
}
}

// Demo function to showcase basic display features
void demo() {
    // Fill screen with a solid color
    tft.fillScreen(DISPLAY_COLOR_BLUE);
    delay(500);

    // Draw some shapes
    tft.fillScreen(DISPLAY_COLOR_BLACK);
    tft.fillRect(20, 20, 100, 50, DISPLAY_COLOR_RED); // Filled rectangle
    tft.drawRect(140, 20, 80, 50, DISPLAY_COLOR_GREEN); // Outline rectangle
    tft.fillCircle(60, 150, 30, DISPLAY_COLOR_YELLOW); // Filled circle
    tft.drawCircle(180, 150, 30, DISPLAY_COLOR_MAGENTA); // Outline circle
    delay(1500);

    // Display text
    tft.fillScreen(DISPLAY_COLOR_BLACK);
    tft.setCursor(10, 10);
    tft.setTextColor(DISPLAY_COLOR_WHITE);
    tft.setTextSize(2);
    tft.println("ST7789 Demo!");
    tft.setCursor(10, 50);
    tft.setTextColor(DISPLAY_COLOR_CYAN);
    tft.setTextSize(1);
    tft.println("Testing shapes, colors, and text...");
    delay(2000);

    // Gradient effect
    for (int i = 0; i < 320; i += 10) {
        tft.fillRect(0, i, 240, 10, tft.color565(i % 255, (i * 2) % 255, (i * 3) %
255));
    }
    delay(2000);
}

```

